

Packet Tracer - Propagate a Default Route in OSPFv2 (Instructor Version)

Instructor Note: Red font color or gray highlights indicate text that appears in the instructor copy only.

Addressing Table

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
R1	G0/0	172.16.1.1	255.255.255.0	N/A
	S0/0/0	172.16.3.1	255.255.255.252	
	S0/0/1	192.168.10.5	255.255.255.252	
R2	G0/0	172.16.2.1	255.255.255.0	N/A
	S0/0/0	172.16.3.2	255.255.255.252	
	S0/0/1	192.168.10.9	255.255.255.252	
	S0/1/0	209.165.200.225	255.255.255.224	
R3	G0/0	192.168.1.1	255.255.255.0	N/A
	S0/0/0	192.168.10.6	255.255.255.252	N/A
	S0/0/1	192.168.10.10	255.255.255.252	
PC1	NIC	172.16.1.2	255.255.255.0	172.16.1.1
PC2	NIC	172.16.2.2	255.255.255.0	172.16.2.1
PC3	NIC	192.168.1.2	255.255.255.0	192.168.1.1
Web Server	NIC	64.100.1.2	255.255.255.0	64.100.1.1

Objectives

Part 1: Propagate a Default Route

Part 2: Verify Connectivity

Background

In this activity, you will configure an IPv4 default route to the Internet and propagate that default route to other OSPF routers. You will then verify the default route is in downstream routing tables and that hosts can now access a web server on the Internet.

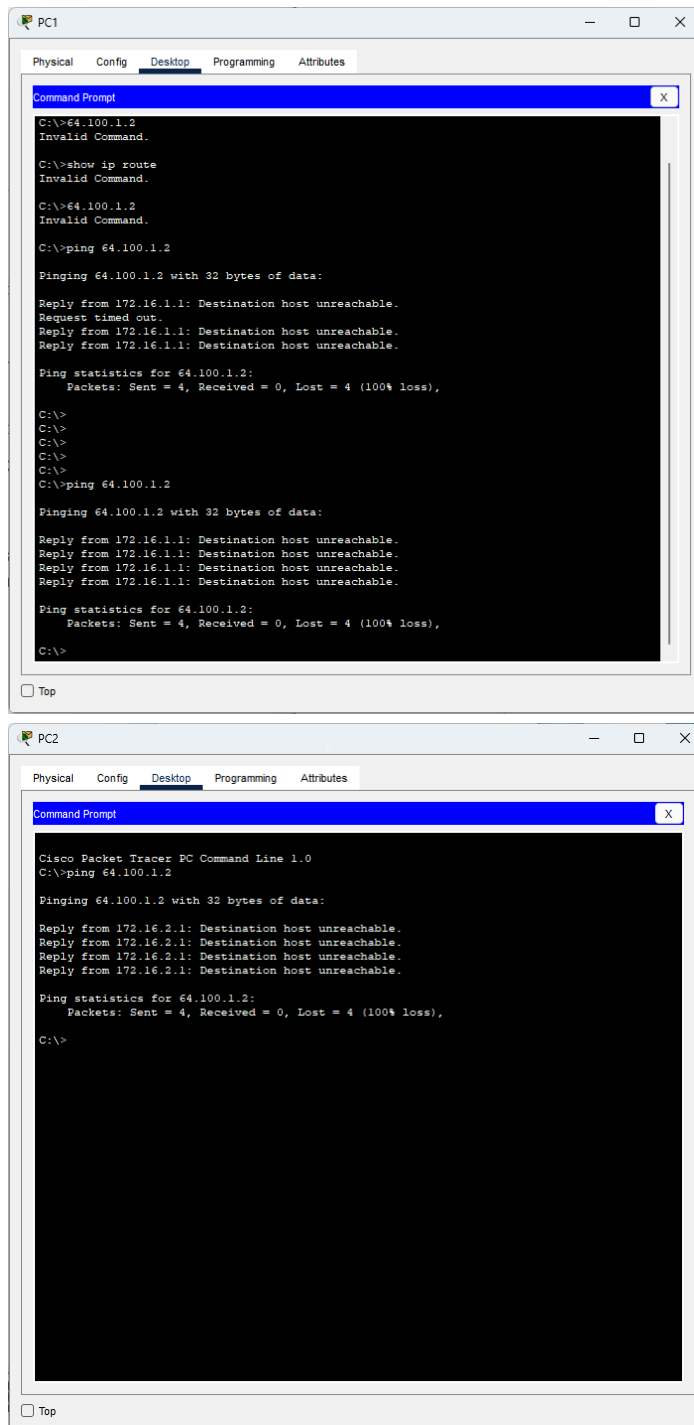
Instructions

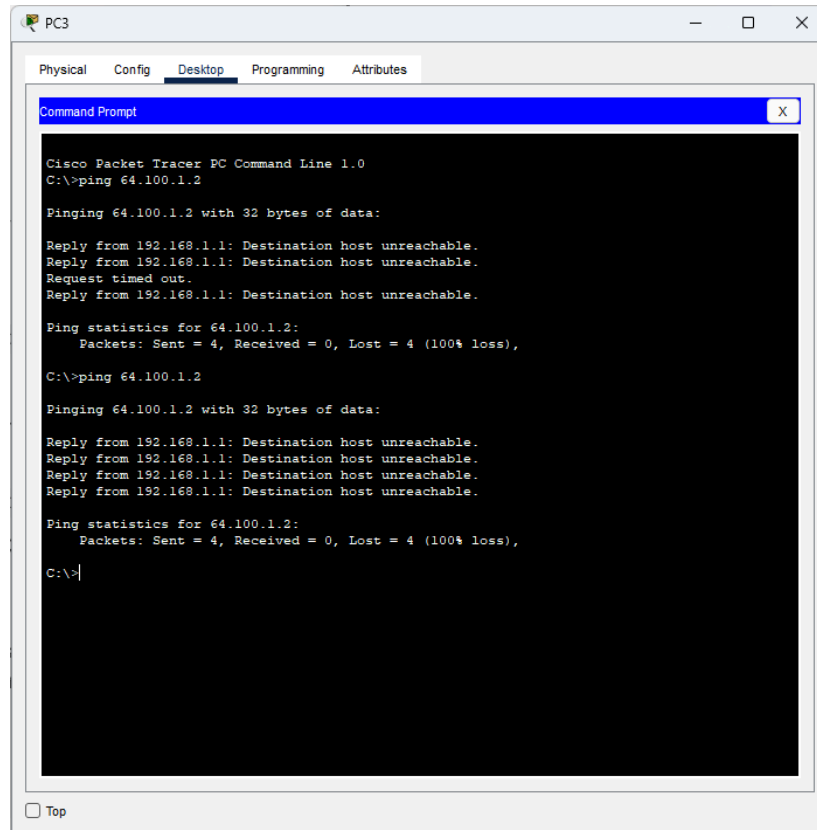
Part 1: Propagate a Default Route

Step 1: Test connectivity to the Web Server

- a. From PC1, PC2, and PC3, attempt to ping the Web Server IP address, 64.100.1.2.

Were any of the pings successful?





No. เมื่อทดสอบ ping จากทั้ง PC1, PC2 และ PC3 ไปยังหมายเลข IP ของ Web Server (64.100.1.2) ผลลัพธ์คือล้มเหลวทั้งหมด แพ็กเก็ตข้อมูลไม่สามารถเดินทางผ่านเครือข่ายออกไปถึง Web Server ปลายทางได้

What message did you receive, and which device issued the message?

Destination unreachable from R2 เมื่อ PC ส่งแพ็กเก็ตเพื่อจะไปหา Web Server ข้อมูลจะถูกส่งผ่าน R1/R3 มาจนถึง Router R2 ซึ่งเป็นเราเตอร์ขอบเขต (Edge Router) แต่เนื่องจาก R2 ยังไม่มีข้อมูลเส้นทางไปหา Web Server มันจึงทิ้งแพ็กเก็ตนั้น แล้วส่งข้อความตอบกลับมานอก PC ดันทางว่า "ไม่พบเส้นทางไปถึงปลายทาง"

- b. Examine the routing tables on routers R1, R2, and R3.

What statement is present in the routing tables that indicates that the pings to the Web Server will fail?

Gateway of last resort is not set โดยปกติ หากเราเตอร์ได้รับแพ็กเก็ตที่ระบุปลายทางเป็นเครือข่ายภายนอกที่ไม่มีอยู่ในตารางเส้นทาง (เช่น IP 64.100.1.2 ของ Web Server) เราเตอร์จะมองหา Default Route เพื่อโยนแพ็กเก็ตนั้นออกไปสู่อินเทอร์เน็ต แต่เนื่องจากยังไม่มีการตั้งค่าส่วนนี้ไว้ ทั้ง R1, R2 และ R3 จึงไม่รู้ว่าจะต้องส่งแพ็กเก็ตต่อไปที่ไหน ทำให้การ ping ไปยัง Web Server ล้มเหลว

Step 2: Configure a default route on R2.

Configure R2 with a directly attached default route to the Internet.

```
R2(config)# ip route 0.0.0.0 0.0.0.0 Serial0/1/0
```

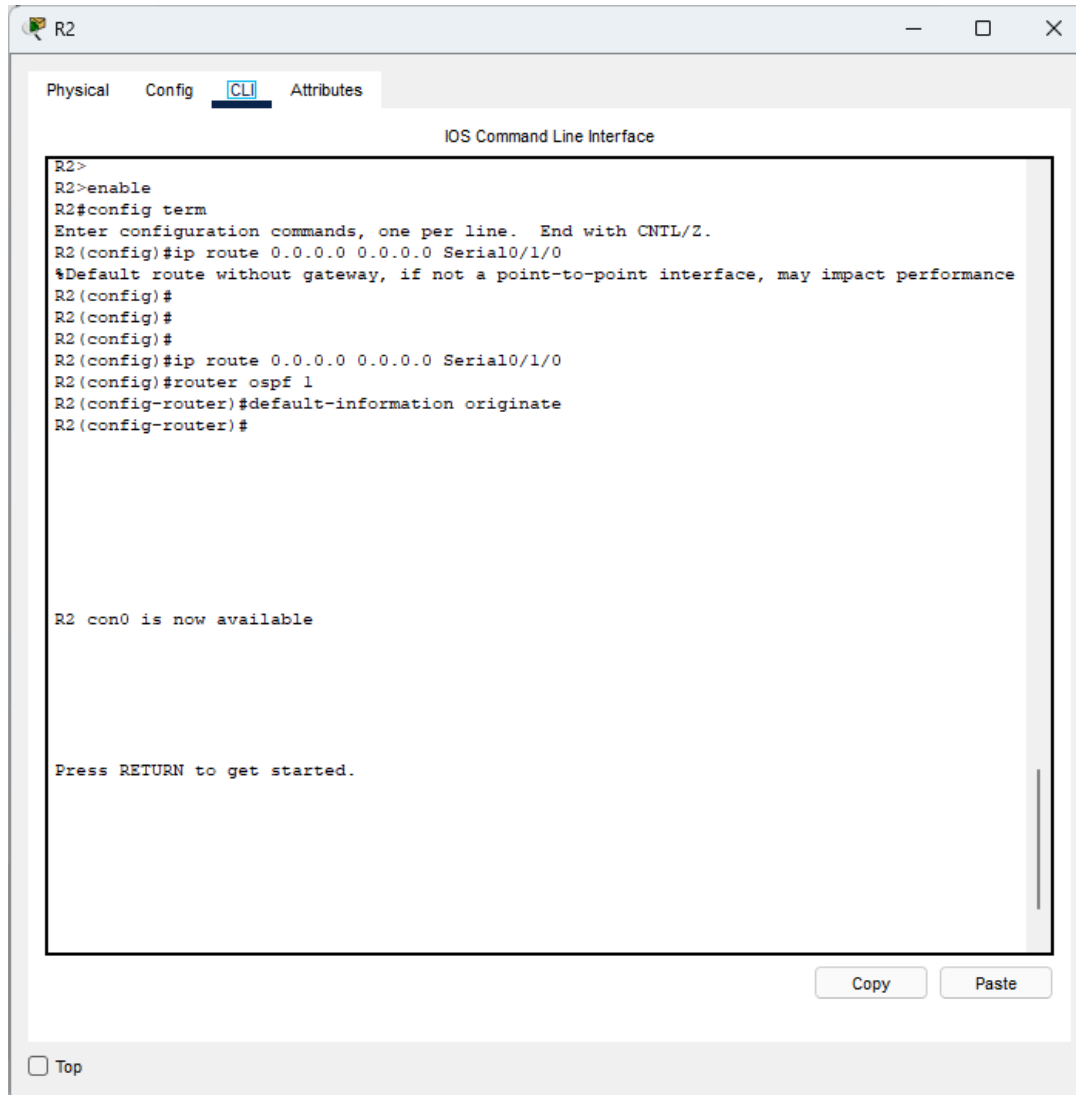
Note: Router will give a warning that if this interface is not a point-to-point connection, it may impact performance. You can ignore this warning because it is a point-to-point connection.

Step 3: Propagate the route in OSPF.

Configure OSPF to propagate the default route in OSPF routing updates.

```
R2(config)# router ospf 1
```

```
R2(config-router)# default-information originate
```



Step 4: Examine the routing tables on R1 and R3.

Examine the routing tables of **R1** and **R3** to verify that the route has been propagated.

```
R1> show ip route
```

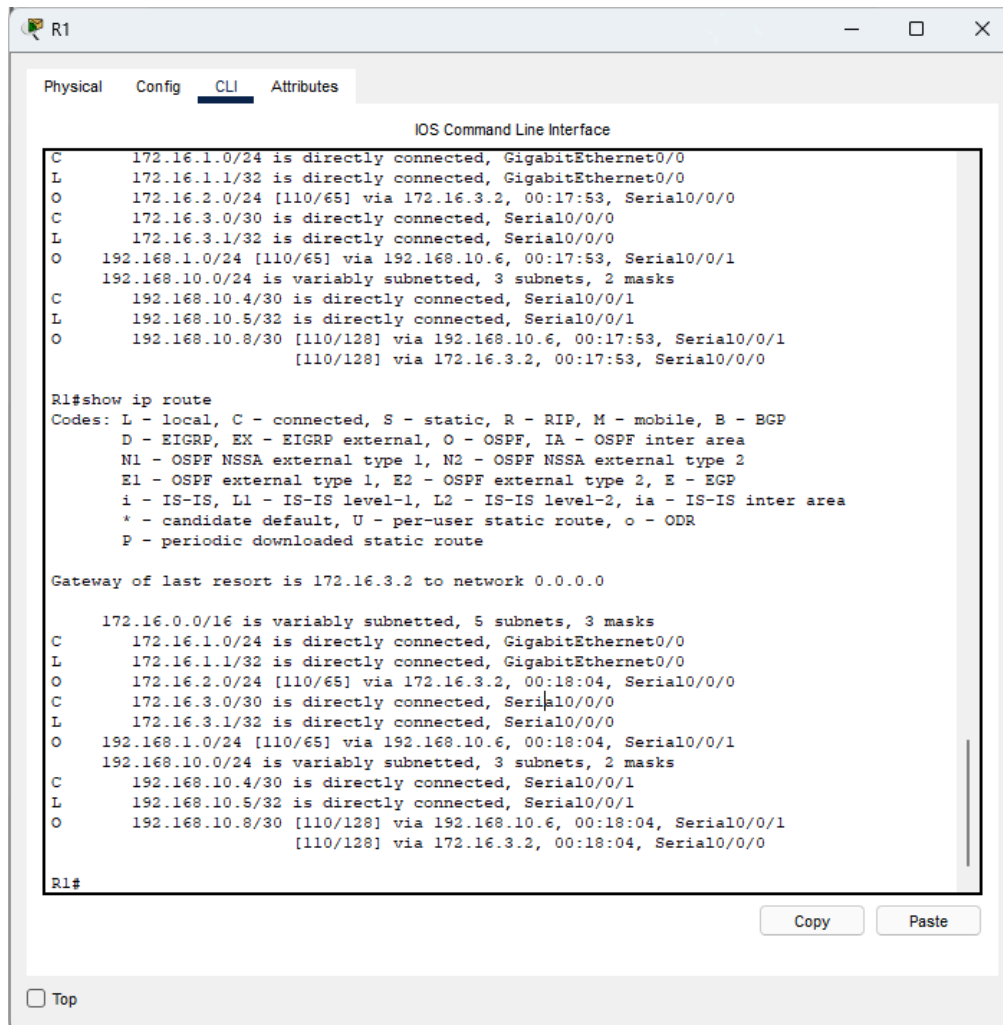
```
<output omitted>
```

```
Gateway of last resort is 172.16.3.2 to network 0.0.0.0
```

```
<output omitted>
```

```
O*E2 0.0.0.0/0 [110/1] via 172.16.3.2, 00:00:08, Serial0/0/0
```

```
!-----
```



R3> show ip route

<output omitted>

Gateway of last resort is 192.168.10.9 to network 0.0.0.0

<output omitted>

O*E2 0.0.0.0/0 [110/1] via 192.168.10.9, 00:08:15, Serial0/0/1

R3

Physical Config **CLI** Attributes

IOS Command Line Interface

```
C 192.168.1.0/24 is directly connected, GigabitEthernet0/0
L 192.168.1.1/32 is directly connected, GigabitEthernet0/0
  192.168.10.0/24 is variably subnetted, 4 subnets, 2 masks
C 192.168.10.4/30 is directly connected, Serial0/0/0
L 192.168.10.6/32 is directly connected, Serial0/0/0
C 192.168.10.8/30 is directly connected, Serial0/0/1
L 192.168.10.10/32 is directly connected, Serial0/0/1
  209.165.200.0/27 is subnetted, 1 subnets
O   209.165.200.224/27 [110/128] via 192.168.10.9, 00:10:23, Serial0/0/1

R3>
R3>enable
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.10.9 to network 0.0.0.0

  172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
O   172.16.1.0/24 [110/65] via 192.168.10.5, 00:20:24, Serial0/0/0
O   172.16.2.0/24 [110/65] via 192.168.10.9, 00:20:24, Serial0/0/1
O   172.16.3.0/30 [110/128] via 192.168.10.9, 00:20:24, Serial0/0/1
     [110/128] via 192.168.10.5, 00:20:24, Serial0/0/0
  192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.1.0/24 is directly connected, GigabitEthernet0/0
L   192.168.1.1/32 is directly connected, GigabitEthernet0/0
  192.168.10.0/24 is variably subnetted, 4 subnets, 2 masks
C   192.168.10.4/30 is directly connected, Serial0/0/0
L   192.168.10.6/32 is directly connected, Serial0/0/0
C   192.168.10.8/30 is directly connected, Serial0/0/1

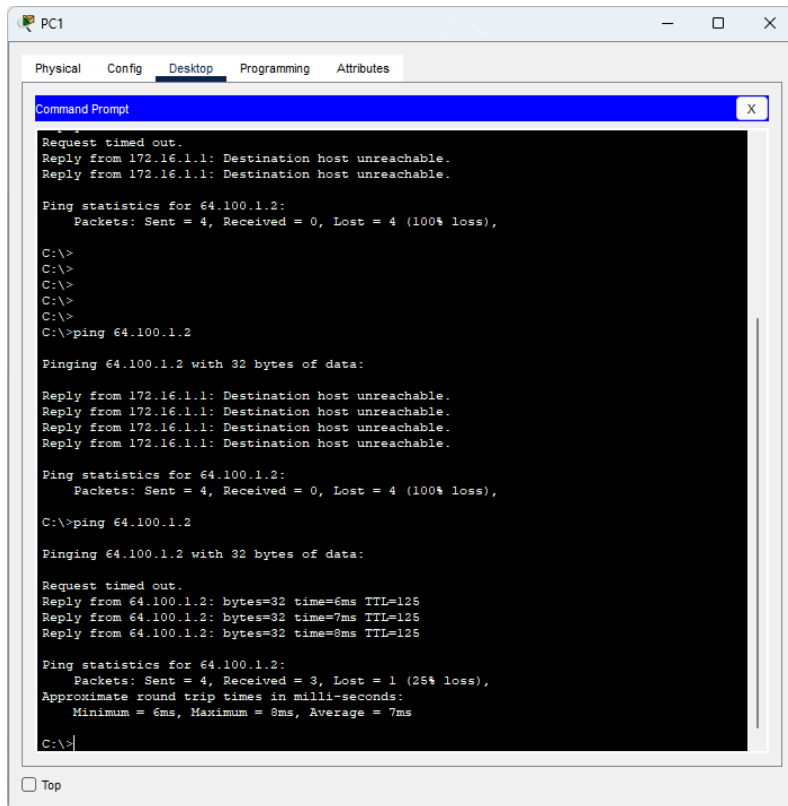
R3#
```

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☐ Top

Part 2: Verify Connectivity

Verify that **PC1**, **PC2**, and **PC3** can ping the web server.



PC1 Command Prompt output:

```
Request timed out.
Reply from 172.16.1.1: Destination host unreachable.
Reply from 172.16.1.1: Destination host unreachable.

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
C:\>
C:\>
C:\>
C:\>
C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Reply from 172.16.1.1: Destination host unreachable.
Reply from 172.16.1.1: Destination host unreachable.
Reply from 172.16.1.1: Destination host unreachable.
Reply from 172.16.1.1: Destination host unreachable.

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

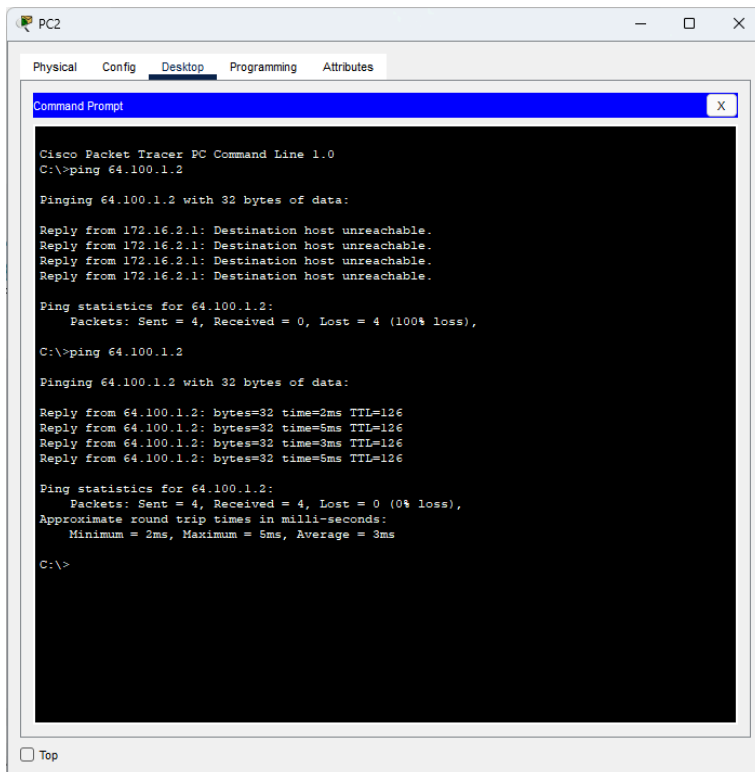
C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Request timed out.
Reply from 64.100.1.2: bytes=32 time=6ms TTL=125
Reply from 64.100.1.2: bytes=32 time=7ms TTL=125
Reply from 64.100.1.2: bytes=32 time=8ms TTL=125

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 8ms, Average = 7ms

C:\>
```



PC2 Command Prompt output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Reply from 172.16.2.1: Destination host unreachable.
Reply from 172.16.2.1: Destination host unreachable.
Reply from 172.16.2.1: Destination host unreachable.
Reply from 172.16.2.1: Destination host unreachable.

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

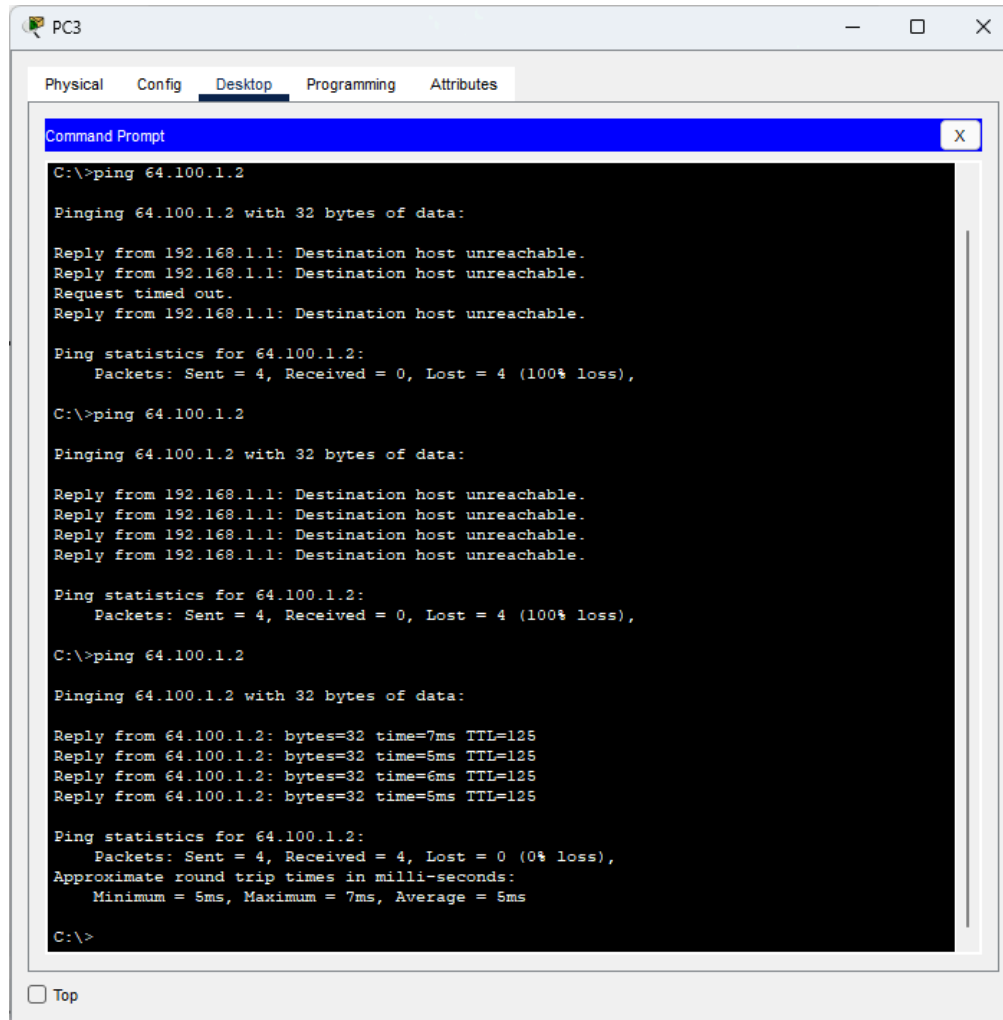
C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Reply from 64.100.1.2: bytes=32 time=2ms TTL=126
Reply from 64.100.1.2: bytes=32 time=5ms TTL=126
Reply from 64.100.1.2: bytes=32 time=3ms TTL=126
Reply from 64.100.1.2: bytes=32 time=5ms TTL=126

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 5ms, Average = 3ms

C:\>
```



Answer script

Router R2

```
enable
config terminal
ip route 0.0.0.0 0.0.0.0 Serial0/1/0
router ospf 1
default-information originate
```

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